

P1_N6_v2 (Room 19): the $\operatorname{Re}(x) \leq 0$
crossing-region estimate – identified as the SAME
ray lemma as Room 17/18, not a separate
construction

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**Question resolved first: is this the ray lemma again,
or the separate part-(ii) machinery?**

Both, and they are not really separable. Re-reading `lem:qi-bounds`
in `paper2a.tex` (lines 2204–2264) line by line settles the ambiguity the task
raised. Part (ii) (line 2213) is:

$$\operatorname{Re} x \leq 0, d'(x) > 0 \implies \log |f(x)| \leq \Upsilon(x)/r + M'(x) + \frac{1}{2} \log \frac{2r}{\pi} + 4 \log(3 + 3e^{-\pi^2 \cos \theta/r}) + K'(x)r,$$

and its proof (line 2249 ff.) has exactly two ingredients:

1. the Jacobi-triple-product identity $(z; P)_\infty = \vartheta(z)/(P/z; P)_\infty$ plus Poisson summation on ϑ , giving the $\Upsilon(x)/r$ landscape term and reducing the sum over $k \in \mathbb{Z}$ to (at most) its two largest terms via a concave-quadratic-in- k argument (the “ $3 + 3e^{-\pi^2 \cos \theta/r}$ ” domination) – this is the piece `P1_N6_note.tex` correctly calls fully N -generic;
2. bounding the surviving factor $(P/z_j; P)_\infty$ – *exactly the same object as in part (i)*, a Pochhammer product along a ray – by **the same ray lemma** proved once, at the top of the proof (line 2228), with $\delta \geq d'(x)$ instead of $\delta \geq d(x)$.

So the answer to the task’s clarifying question 1 is: **it is the same ray-lemma analysis as Room 17/18, evaluated at a different sign of the argument a' , wrapped inside a separate (but N -generic) book-keeping layer (ingredient 1).** There is no second, independent “part-(ii)

construction” distinct from the ray lemma; `P1_N6_note.tex`’s framing of part (ii) as needing its own generalisation was really pointing at the same ray-lemma degeneracy problem Room 16/17/18 already dug into for part (i), just not phrased that way at the time.

The key algebraic fact that resolves the task’s point 2

The two separation constants are *not* both $\mathcal{D}(2 \operatorname{Re} x, \cdot)$. Reading the definitions at lines 2196–2197:

$$d(x) = \min_{c^N=1} \mathcal{D}(2 \operatorname{Re} x, \cdot) \quad (\text{part (i), } \operatorname{Re} x \geq 0), \quad d'(x) = \min_{c^N=1} \mathcal{D}(-2 \operatorname{Re} x, \cdot) \quad (\text{part (ii), } \operatorname{Re} x \leq 0)$$

The sign of the first argument of $\mathcal{D}$ flips between the two parts. This is not a bookkeeping accident: it is exactly what keeps $a' \geq 0$ in both regimes. In part (i), $\operatorname{Re} x \geq 0 \Rightarrow a' = 2 \operatorname{Re} x \geq 0$. In part (ii), $\operatorname{Re} x \leq 0 \Rightarrow a' = -2 \operatorname{Re} x \geq 0$. **So the worry raised in the task – that $a' = 2 \operatorname{Re}(x)$ might go negative or make $1 - e^{-a'}$ vacuous on $\operatorname{Re}(x) \leq 0$ – does not arise, because that is not the quantity part (ii) actually uses.** The paper’s own construction already routes around this by defining a second constant with the opposite sign convention. This was implicit in `lem:qi-bounds` itself and is independent of N : it is visible already at $N = 4$, and nothing about it privileges $N = 4$ over $N = 6$.

Consequence: Room 17/18’s bound already secures the ray-lemma sub-piece of the crossing region

Room 17/18’s proof (`room17_consolidated.tex`) is: for $Y(\rho) = 1 - e^{-a' - \rho \cos \theta}$, $Y'(\rho) = \cos \theta e^{-a' - \rho \cos \theta} \geq 0$ whenever $\cos \theta \geq 0$, *for any real a'* , so Y is non-decreasing and $\mathcal{D}(a', \varphi) \geq Y(0) = 1 - e^{-a'}$. The proof never uses $a' \geq 0$; it only needs $\cos \theta \geq 0$. What makes the bound *useful* (positive) is $a' \geq 0$, and that is supplied automatically by d' ’s sign convention on $\operatorname{Re} x \leq 0$ exactly as it was supplied for d on $\operatorname{Re} x \geq 0$.

At $N = 6$ ’s certified saddle angle $\theta^* = 0$ (so $\cos \theta^* = 1 \geq 0$, the best case for this bound, as already noted in Room 17), this gives, for every x with $\operatorname{Re} x \leq 0$ on the relevant contours:

$$d'(x) \geq 1 - e^{2 \operatorname{Re}(x)} > 0,$$

uniformly over all 6 resonance classes $c^6 = 1$ simultaneously (the bound never needs to know which class is closest to resonance, exactly as Room 17 found for $d(x)$), and uniformly over $\text{Im}(x)$, since the bound depends only on $\text{Re}(x)$ once $\cos \theta \geq 0$ is fixed. This directly resolves the *Hilbert* concern recorded in `P1_N6_note.tex` (“five codimension-1 loci $\arg(\zeta_6^i) - 2\text{Im}(x) \equiv 0$ where the angular safety net vanishes”): those loci are where the *angular* term $\sigma(\varphi - \rho \sin \theta)$ degenerates, but Room 17/18’s bound never invokes the angular term at all – it is a pure-modulus bound, $\max\{\sigma(\cdot), Y(\rho)\} \geq Y(\rho)$ discards the angular piece entirely. So the five loci are not actually dangerous for *this* estimate; they would only matter for a bound that tried to extract the sharper, angle-dependent value of $d'(x)$ (as Hilbert’s 0.866 figure did at the single slice $\theta = 0, \rho = 0$), which is not needed here since a genuine lower bound, not the exact value, is what part (ii)’s error term $K'(x)$ needs (exactly as Room 17 already noted for part (i)’s analogous term).

What this does and does not close

Secured (this room): the ray-lemma sub-piece of `lem:qi-bounds(ii)` – the bound $\delta \geq d'(x)$ on the surviving Pochhammer factor $(P/z_j; P)_\infty$ – transfers to $N = 6$ at $\theta^* = 0$ by the identical argument as Room 17/18, with no sign obstruction and no per-resonance-class case analysis needed, because (a) part (ii) uses $a' = -2\text{Re } x \geq 0$ on its own domain, not $2\text{Re } x$, and (b) Room 17/18’s bound is angle-independent once $\cos \theta \geq 0$. This is a genuine, if narrow, positive extension: it answers “does Room 17/18’s fix survive into the crossing region” with **yes, by construction of the paper’s own sign convention, not by any new argument needed here.**

NOT secured (still open, unchanged from `P1_N6_note.tex`): the *other* ingredient of part (ii), the Jacobi-triple-product/Poisson-summation bookkeeping that produces $\Upsilon(x)$, $M'(x)$, $K'(x)$ with their explicit constants $(\frac{1}{\sqrt{2}} + 2, \frac{\pi}{2}, \frac{73}{12}, \frac{4}{3}, \dots)$ is asserted but not re-derived for $N = 6$ anywhere in this repository. `P1_N6_note.tex`’s own weak point (§??, item 2 there) already flagged this honestly as unverified, and nothing in this room changes that: the mod-4 residue sum $\sum_{\varrho=0}^3 \max_{m \equiv \varrho(4)} \Upsilon(x)$ (line 2190) needs its literal mod-6 analogue $\sum_{\varrho=0}^5 \max_{m \equiv \varrho(6)}$ written out, the concave-quadratic domination constant “ $3 + 3e^{-\pi^2 \cos \theta / r}$ ” needs rechecking for the coarser mod-6 lattice (the domination argument is structurally N -generic – a concave quadratic in k is dominated by its two largest terms regard-

less of the lattice spacing – but the numeric constant 3 came from the specific mod-4 tail sum and was not rederived for mod-6 here), and the explicit numeric constants inside $M'(x), K'(x)$ (which come from finitely many termwise Euler–Maclaurin/ray-lemma error estimates specific to the 4-term product $\prod_{j=1}^4 (i^j p^{4s+j}; P)_\infty$) need their 6-term analogue $\prod_{j=1}^6 (\zeta_6^j p^{6s+j}; P)_\infty$ worked out from scratch. This is exactly the “mechanical but unverified” bookkeeping step, not the ray-lemma obstruction, and nothing found in this room closes it.

Verdict

CLARIFIED SCOPE, with one sub-piece SECURED. Precisely:

- The task’s ambiguity (ray lemma extended vs. separate part-(ii) construction) is resolved: part (ii) is not a separate construction, it is the same ray lemma applied to the same kind of Pochhammer factor with the opposite-signed argument $a' = -2 \operatorname{Re} x$, wrapped in an N -generic Poisson-summation layer.
- The specific danger flagged in the task (Room 17’s bound going vacuous or negative on $\operatorname{Re}(x) \leq 0$) **does not occur**: the paper’s own $d'(x)$ already uses $-2 \operatorname{Re} x \geq 0$ on that domain, so Room 17/18’s monotonicity argument secures $d'(x) \geq 1 - e^{2 \operatorname{Re} x} > 0$ at $N = 6$ ’s $\theta^* = 0$ with no new work, and as a byproduct resolves the “five loci” uniformity worry from `P1_N6_note.tex`’s Room 1 (Hilbert seat), since that worry only threatened the angular term, which this bound never uses.
- The genuinely open remainder for the full $N = 6$ analogue of part (ii) – and hence for the direct tie-ray statement – is now narrowed to a single, concrete, well-scoped task: writing out the mod-6 Poisson-summation/triple-product bookkeeping ($\Upsilon(x), M'(x), K'(x)$ with correct explicit constants for the 6-term product) that `P1_N6_note.tex` asserted was “mechanical” but never carried out. No obstruction to this has been found; it has simply not been attempted, here or elsewhere in the repository.